

Joe Lopez

Seattle, WA

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WORK EXPERIENCE

Microsoft Corporation

Redmond, WA

Software Development Engineer 2

Jul 2023 – May 2025

- Planned and implemented second-gen firmware according to new hardware layout. Developed design plan and distributed work amongst vendor team and oversaw firmware delivery timeline.
- Implemented and future-proofed novel features such as PCIe hot plug according to specifications, adding integration tests to exercise different software interactions without human intervention.
- Inspired cross-team collaboration by facilitating work forum for PCIe/CXL hardware and software stakeholders for design reviews, bug investigations, and tech talks.
- Accelerated development process by leveraging internal tool and agentic AI to generate firmware unit tests for uncovered lines of code using techniques such as RAG and multi-model approaches.

Firmware Development Engineer

Sep 2021 – Jul 2023

- Developed firmware for PCIe hardware initialization on company's first purpose-built ARM server platform containing first-party Neoverse N2 SoC.
- Mitigated failures during chip and platform using Windows Debugger (WinDbg) and ACPI tools, releasing a firmware patch to unblock others.
- Led the integration of firmware unit test suite into platform codebase, consuming open source framework and modifying it for our usecase.
- Collaborated with internal open-source owners to upstream enhancements for consumption by all first-party and third-party UEFI teams.
- Integrated pipeline for unit test CI/CD into platform codebase, capturing code coverage statistics and gating new pull requests.
- Enhanced platform Reliability, Availability, and Serviceability (RAS) by constructing drivers for PCIe error reporting and handling, including workarounds for various silicon errata.

Robert Bosch Tool Corporation (Dremel)

Mount Prospect, IL

Electrical Engineering Intern

May 2019 – Aug 2019

- Increased data acquisition capacity by rewriting tool firmware for custom logger

EDUCATION

University of Illinois

Urbana-Champaign, IL

Bachelor's in Computer Engineering

Aug 2017 – Jun 2021

LEADERSHIP EXPERIENCE

Illini Formula Electric

Data Acquisition Lead

Aug 2019 – Aug 2020

- Led a team to design and implement a data logger for vehicle sensor data.

Theta Tau Professional Engineering Fraternity

Projects Committee Chair

Aug 2019 – Jun 2020

- Managed a committee of 18 members to develop a project for presentation

SKILLS

Programming: C, C++, UEFI Framework, Python, CI/CD, Git, GDB, TRACE32, WinDBG

Embedded Electronics: ARM, PCIe, STM32, I2C, SPI, UART, CAN